

Adarsh Dattarao Raut

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📄 https://scholar.google.com/citations?hl=en • 🆔 0009-0001-6554-4709

Education

Savitribai Phule Pune University (SPPU) **Pune, MH, India**
M.Tech. Atmospheric Sciences , (Joint Programme with IITM Pune) **2024**

Dr. Babasaheb Ambedkar Technological University(BATU) **Raigad, MH, India**
B.Tech. Mechanical Engineering **2021**

Professional Appointments

Project Engineer **Centre for Development of Advanced Computing (C-DAC) Pune**
Nov/2025–Present

Contribute to operational WRF-based Numerical Weather Prediction (NWP) workflows on High-Performance Computing (HPC) systems. Develop and maintain Python-based data processing pipelines for large atmospheric datasets (ERA5, IMD, NetCDF, GRIB). Perform high-resolution climate diagnostics, extreme rainfall analysis, and urban climate risk assessment using meteorological and socio-economic datasets. Utilize parallel data-processing techniques (Dask, multiprocessing, Xarray) for efficient handling of large-scale climate datasets. Support model execution, post-processing, visualization, and verification of weather and climate simulations.

Project Associate-I **Indian Institute of Tropical meteorology (IITM) Pune**
May/2025–Sept/2025

Investigated large-scale drivers of Indian monsoon variability using observational, reanalysis, and seasonal prediction datasets. Analysed outputs from the Monsoon Mission Climate Forecast System Version 2 (MMCFSv2) for seasonal forecasting applications. Developed Python-based workflows for processing and analysing multi-terabyte climate datasets. Applied statistical and diagnostic techniques to understand atmosphere-ocean teleconnections and monsoon predictability.

Project Associate-I **National Institute of Oceanography (NIO)Goa**
Aug/2024–May/2025

Conducted Computational Fluid Dynamics (CFD) simulations of oscillating and flapping foil propulsion systems using ANSYS Fluent. Performed numerical investigations of hydrodynamic performance, thrust generation, and flow-structure interactions. Assisted in validation of numerical simulations against experimental observations. Processed and analysed large simulation datasets using scientific computing tools. Contributed to scientific reports, conference presentations, and manuscript preparation.

Graduate Research Assistant **IITM Pune**
M. Tech Thesis: Arctic Oscillation Influence on Indian Summer Monsoon Rainfall **2023–2024**

Computed the Arctic Oscillation (AO) Index from reanalysis datasets using EOF/PCA techniques. Analysed climate teleconnections using ERA5, NCEP/NCAR, and rainfall datasets. Developed reproducible Python workflows for climate data analysis, visualization, and statistical modelling. Processed large-scale atmospheric datasets in Linux-based scientific computing environments.

Drafting Engineer **Kakade Laser Pune**
CFD and CAD Modeling **2021–2022**

Publications

1. Muchhala D., Maurya P.K., **Adarsh Raut.**, et al. Bioinspired Underwater Robotics: Advances Across the Materials, Design, Control, and Applications. Robotics and Autonomous Systems, Elsevier, 2026..

Conference Presentations

Adarsh Raut, Maurya P., Muchhala D.

Impact of Pitching Angle and Frequency on Flapping Foil Hydrodynamics: A CFD Study with Experimental Validation

ICSEE 2025

- CFD simulation of oscillating airfoil/flapping foil dynamics
- Investigation of lift and thrust generation under varying pitching frequencies and amplitudes
- Experimental validation showing strong correlation between CFD and measured thrust

Skills

- **HPC & Scientific Computing:** Linux/Unix, HPC Clusters, Slurm, Bash Scripting, Dask, Multiprocessing, Xarray, NetCDF4, Git/GitLab, WRF/WPS Installation & Compilation, Scientific Software Development, Parallel Climate Data Processing.
- **Programming Languages:** Python, Fortran 90/95, C, MATLAB, R.
- **Climate & Data Analysis Tools:** Climate Data Operators (CDO), Xarray, Pandas, NumPy, NetCDF, GRIB, Cartopy, Matplotlib, Panoply, Ncview.
- **Numerical Modelling:** Weather Research and Forecasting (WRF), Computational Fluid Dynamics (CFD), Boundary Layer Modelling, Atmospheric Diagnostics, Statistical Analysis.

Key Projects

- **Operational WRF-based Numerical Weather Prediction and Climate Diagnostics (2026-Present)**
 - Supported operational Weather Research and Forecasting (WRF) model workflows on HPC systems.
 - Processed and analysed large meteorological datasets including ERA5 and IMD observations.
 - Developed automated post-processing and visualization pipelines using Python and scientific computing tools.
 - Conducted extreme rainfall diagnostics and urban climate risk assessment studies.
- **High-Performance Processing of Atmospheric Datasets (2024-2025)**
 - Developed parallel workflows using Dask, Multiprocessing, and Xarray for large climate datasets.
 - Worked extensively with NetCDF, GRIB, ERA5, IMD, NCEP/NCAR, and WRF model outputs.
 - Optimized data processing pipelines for efficient analysis on Linux-based HPC environments.
 - Applied reproducible scientific computing practices using Git and version-controlled workflows.
- **Seasonal Prediction and Climate Model Data Analysis using MMCFSv2 (2023-2024)**
 - Analysed Monsoon Mission Climate Forecast System Version 2 (MMCFSv2) outputs for seasonal forecasting applications.
 - Processed multi-terabyte climate model datasets using Python, Xarray, and parallel data analysis techniques.
 - Investigated large-scale monsoon drivers and atmosphere-ocean teleconnections.
 - Contributed to climate diagnostics and model evaluation workflows.
- **Arctic Oscillation Index Computation and Teleconnection Analysis (2022-2023)**
 - Computed the Arctic Oscillation (AO) Index from NCEP/NCAR and ERA5 reanalysis datasets using EOF/PCA techniques.

- Developed Python-based workflows for large-scale climate data processing and statistical analysis.
- Investigated the influence of AO variability on Indian Summer Monsoon Rainfall using observational and reanalysis datasets.
- Utilized NetCDF, Xarray, NumPy, and HPC resources for efficient climate data analysis.

Selected Training, Workshops & Professional Development

- Apr 2025: National Training Workshop on Scientific Writing, Indian Institute of Tropical Meteorology (IITM), Pune
- Nov 2023: Advanced WRF Modeling Workshop, Indian Institute of Tropical Meteorology (IITM), Pune
- Jan 2023: Numerical Weather Prediction and WRF Modeling, Atmospheric Science Coursework
- Apr 2023: Statistical Methods for Atmospheric Science, Atmospheric Science Coursework
- Dec 2024: GIS and Remote Sensing Applications for Environmental Analysis
- Oct 2023: Understanding and Visualizing Data with Python, Coursera
- Oct 2022: Boundary Layer Meteorology, Atmospheric Science Coursework
- Sep 2022: Introduction to Atmospheric Science, NPTEL
- Feb 2021: Computational Fluid Dynamics (CFD) Modeling and Simulation

Awards & Scholarships

- Government of India Scholarship (Ministry of Social Justice and Empowerment), awarded throughout B.Tech. (2017–2021) and M.Tech. (2022–2024) studies for academic merit.

Language Skills

English , Hindi , Marathi